

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

1. Motherboard

The primary circuit board that connects all hardware components together, allowing communication between the CPU, RAM, storage, and peripheral devices.

2. Central Processing Unit (CPU / Processor)

The "brain" of the computer, mounted directly on the motherboard socket, which executes instructions and processes system data.

3. CPU Cooler (Heat Sink & Fan / Liquid Cooler)

Sits directly on top of the CPU to dissipate the extreme heat generated during operation and keep processing temperatures safe.

4. Memory (RAM)

High-speed temporary storage sticks inserted into motherboard slots that hold active data and running applications for quick access by the CPU.

5. Graphics Processing Unit (GPU / Expansion Card)

A dedicated expansion card plugged into a PCIe slot that processes complex visual data for displays, gaming, 3D rendering, and video editing.

6. Storage Drives (SSD / HDD)

Permanent data storage devices—such as fast NVMe M.2 SSDs, standard 2.5-inch SATA SSDs, or 3.5-inch Hard Disk Drives (HDDs)—that store the operating system, programs, and files.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



Motherboard

- **Function:** The primary circuit board that connects all PC parts together so they can share information and communicate.

CPU (Processor)

- Function: The "brain" of the computer that handles commands, calculations, and runs programs.

Storage (SSD / Hard Drive)

- Function: Long-term memory where Windows, programs, games, and personal files are permanently saved.

Power Supply Unit (PSU)

- Function: Takes electricity from the wall outlet and converts it into power for all internal parts.

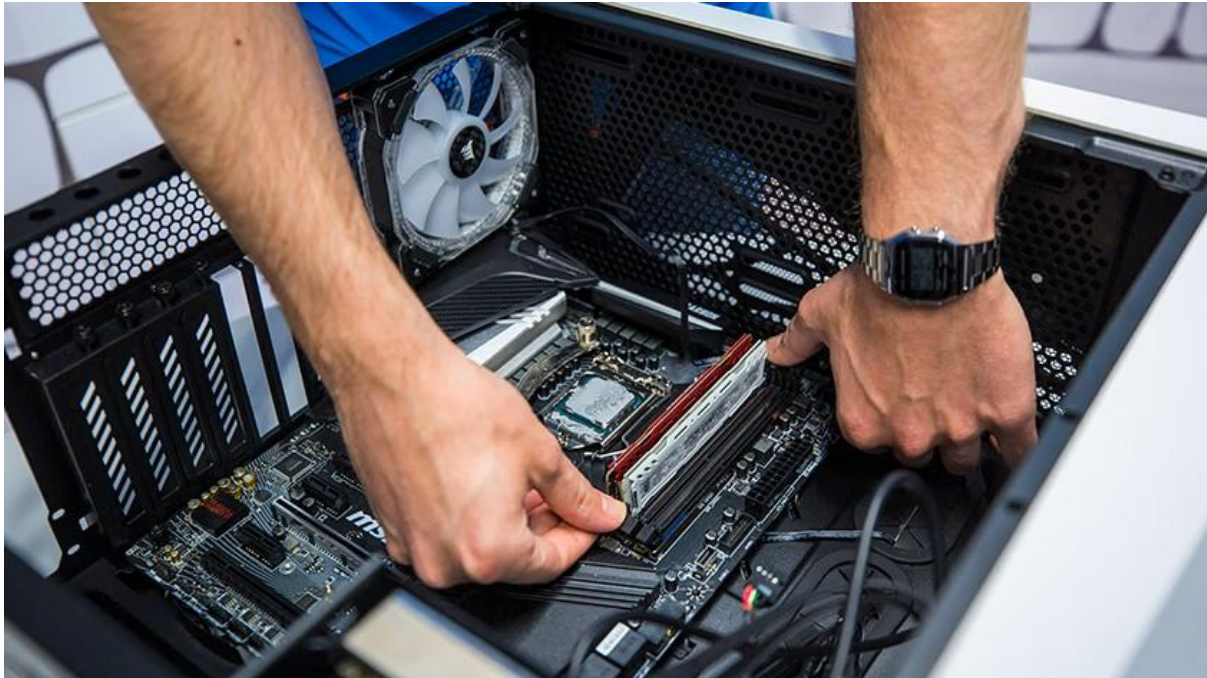
Steps to Disassemble a Desktop PC

Step 1 & 2: Unplugging Power & Opening the Side Panel



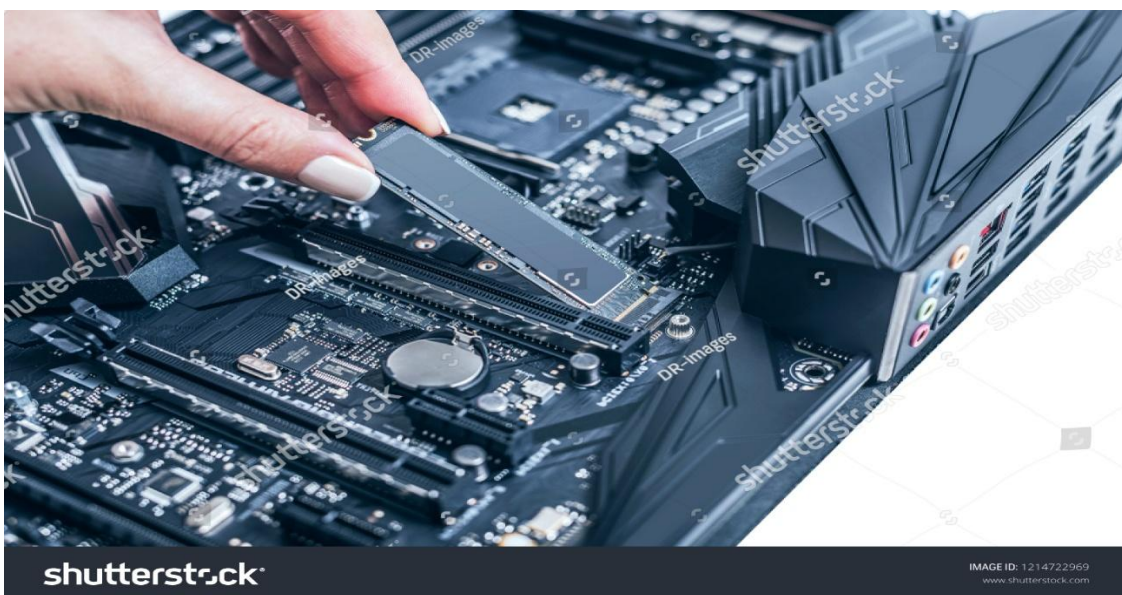
Step 1 & 2: Turned off the PC power supply, disconnected all main power cables, and unscrewed the side panel to open the cabinet.

Step 3 & 4: Removing RAM Modules



Caption: Step 3 & 4: Unplugged the internal power supply wires and pushed open the plastic retention clips to remove the RAM stick from its slot.

Step 5: HDD/SSD, and motherboard.



3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step 1: Install CPU, M.2 SSD, and RAM onto Motherboard

Install the CPU into its socket, secure the retention arm, insert the M.2 NVMe SSD into its slot, and click the RAM sticks firmly into the DIMM slots until the retention tabs lock.

Step 2: Install I/O Shield, Case Fans, and Power Supply Unit (PSU)

Snap the motherboard's rear I/O shield into the back cutout, mount case intake/exhaust fans, and slide the PSU into its bay, securing it with rear screws.

Step 3: Mount Motherboard into Cabinet

Align the motherboard with the pre-installed brass standoffs in the case and secure it using the motherboard standoff screws.

Step 4: Install CPU Cooler & Connect Motherboard Power Cables

Mount the CPU cooler, plug in the CPU_FAN header, and connect the main 24-pin ATX power and 8-pin EPS/CPU power cables from the PSU.

Step 5: Install 2.5"/3.5" Storage Drives (SATA SSD / HDD)

Slide HDDs or 2.5" SATA SSDs into their drive cages, secure them with screws or trays, and connect both their SATA power cables (from PSU) and SATA data cables (to motherboard).

Step 6: Install Graphics Card (GPU) & Connect PCIe Power

Align the GPU with the primary top PCIe x16 slot, push down until the latch clicks, secure the PCIe bracket to the case with screws, and plug in the PCIe power connectors.

Step 7: Connect Front Panel Headers, Cable Management & Close Cabinet

Plug in front panel connectors (Power SW, Reset, Front USB, HD Audio), bundle loose cables behind the motherboard tray with zip ties, slide the side panel back onto the chassis, and plug in external peripherals and main AC power.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Required & Recommended Tools:

Magnetic Phillips Screwdriver

Anti-Static Wrist Strap (ESD Strap)

Non-Conductive Plastic Pry Tools / Spudger

Zip Ties or Hook-and-Loop Cable Straps

Thermal Paste

Parts Tray or Magnetic Screw Bowl

Essential Safety & Preparation Precautions:

1. Power Off & Unplug Completely

- Shut down the OS, flip the switch on the back of the PSU to O (Off), unplug the main power cable from the wall, and press the case power button once or twice to bleed remaining capacitor charge.

2. Work on a Non-Conductive, Hard Surface

- Assemble or disassemble your system on a clean wooden, plastic, or glass desk. Avoid working on carpets, rugs, or using towels underneath the PC.

3. Handle Components Safely by Their Edges

- Hold RAM modules, expansion cards, and motherboards by their outer plastic/PCB edges. Never touch gold contact pins, CPU socket pins, or surface-mounted microcontrollers directly.

4. Remove Jewelry

- Take off metal rings, watches, and bracelets.

5. Never Force Components or Fasteners

- If a RAM stick, GPU, cable, or screw requires extreme force, stop and re-align it. Most PC parts are keyed (notched) to fit in only one orientation.